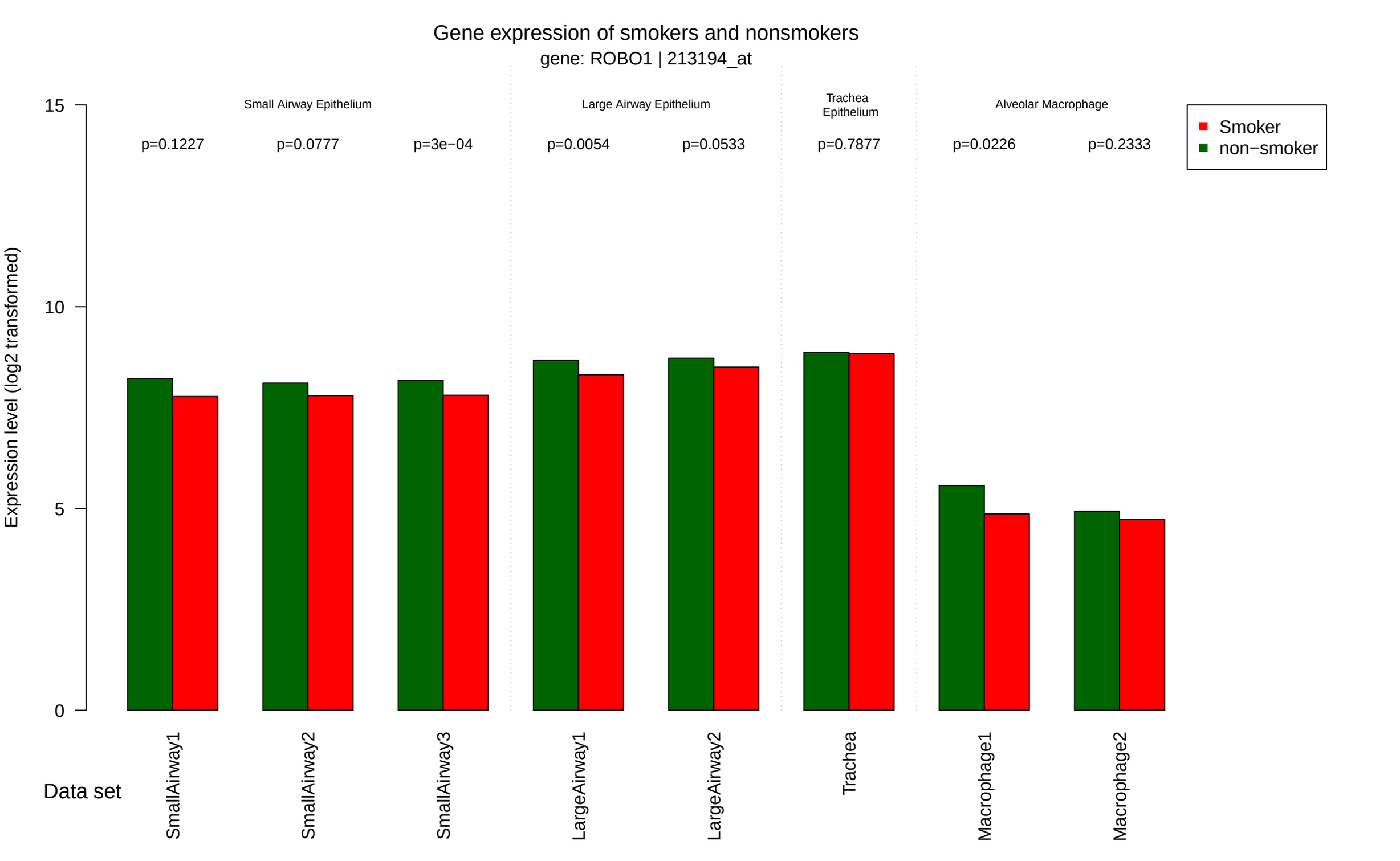
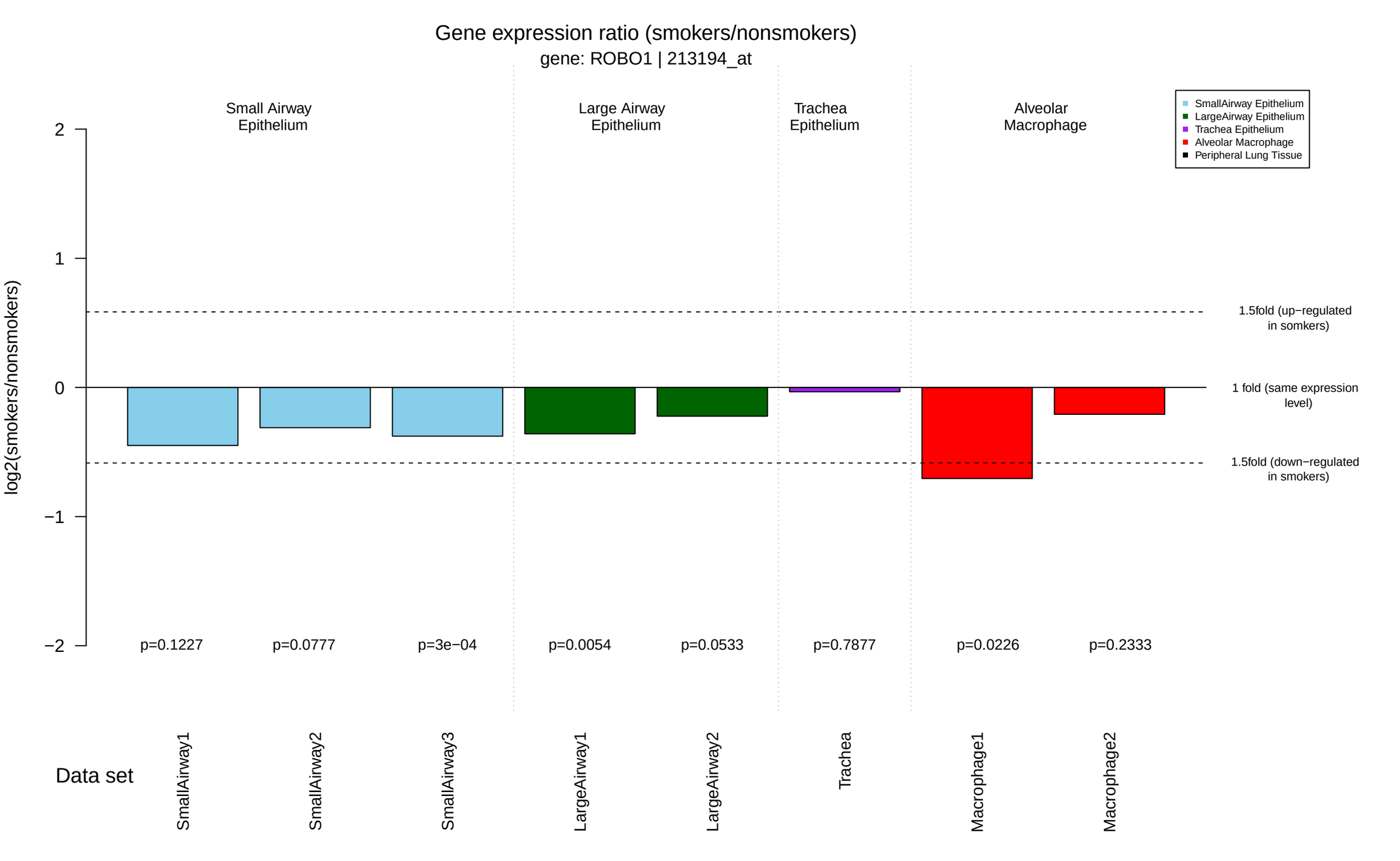
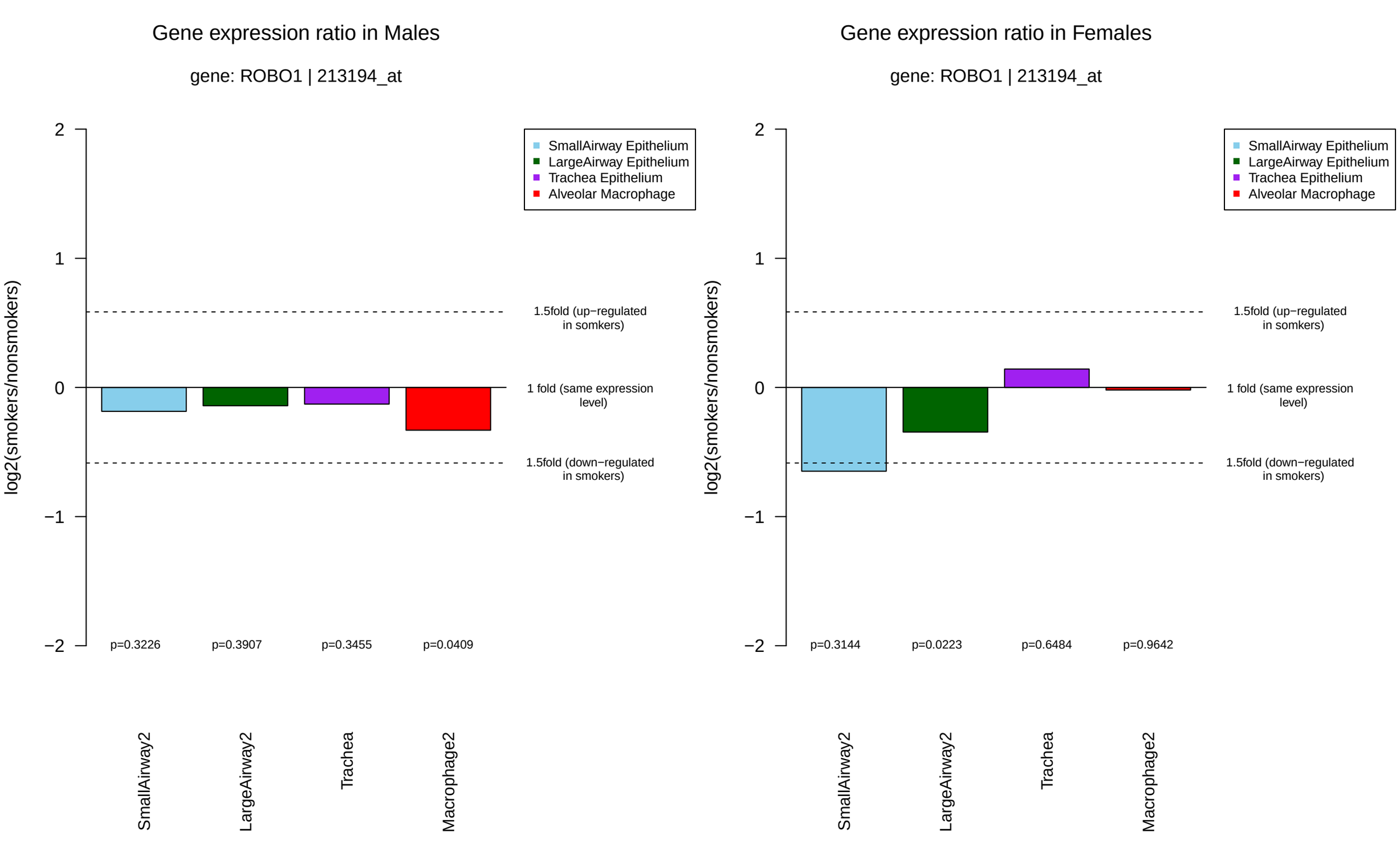
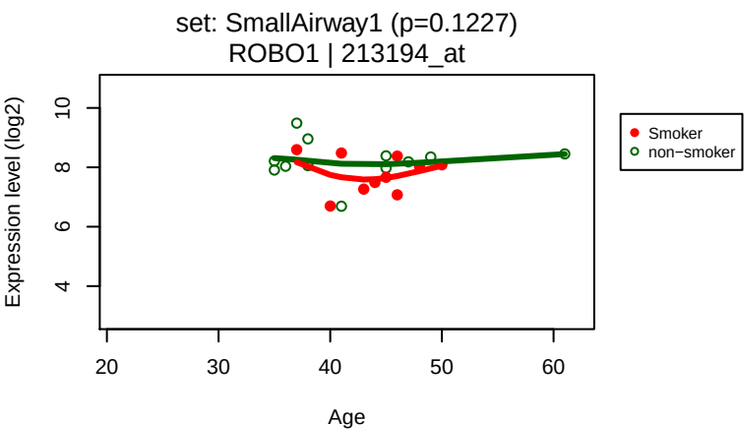
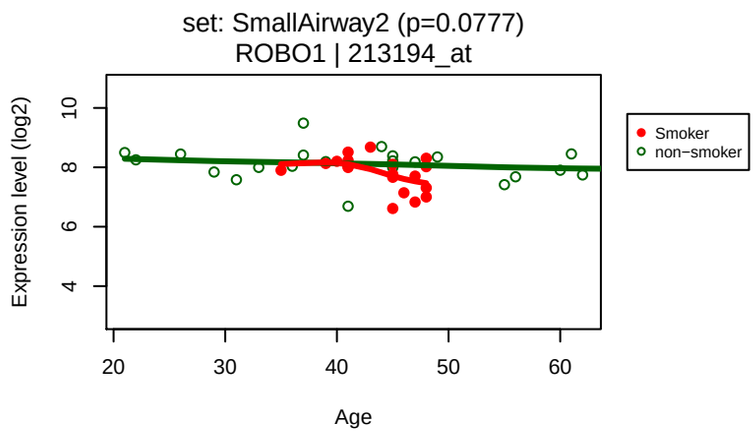
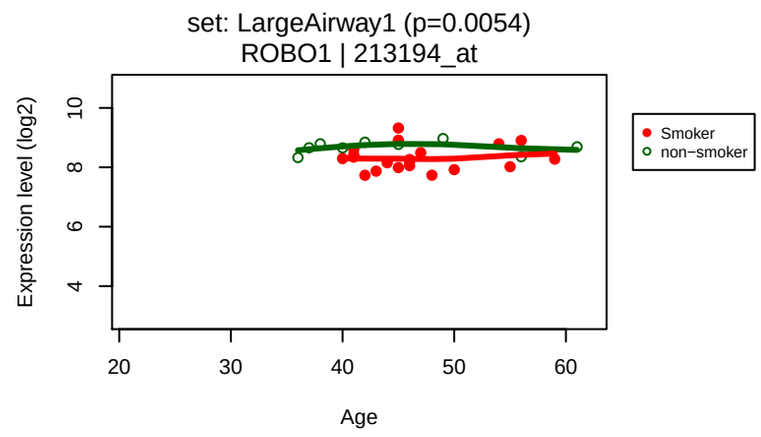

Gene expression of smokers and nonsmokers
gene: ROBO1 | 213194_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1227
p=0.0777
p=3e−04
p=0.0054
p=0.0533
p=0.7877
p=0.0226
p=0.2333
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

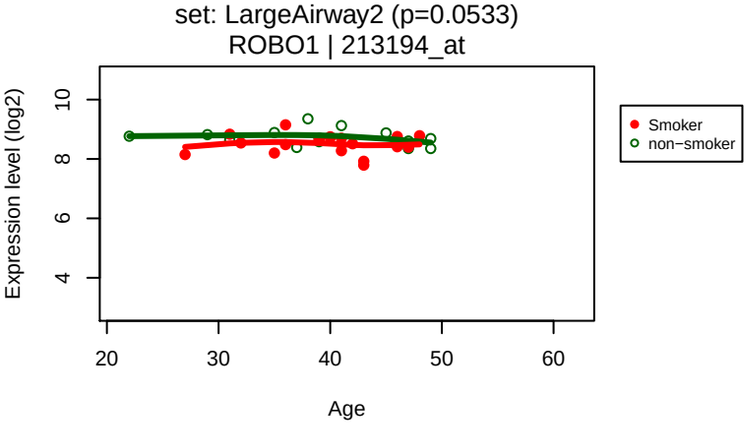

Gene expression ratio (smokers/nonsmokers)
gene: ROBO1 | 213194_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1227
p=0.0777
p=3e−04
p=0.0054
p=0.0533
p=0.7877
p=0.0226
p=0.2333
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

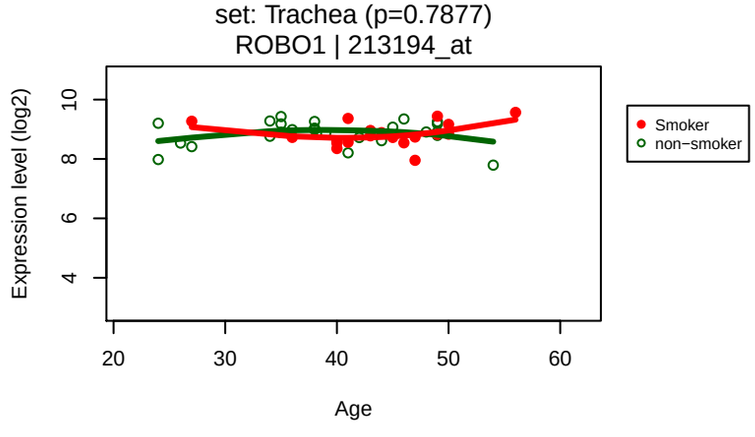

Gene expression ratio in Males
gene: ROBO1 | 213194_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3226
p=0.3907
p=0.3455
p=0.0409
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ROBO1 | 213194_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3144
p=0.0223
p=0.6484
p=0.9642
SmallAirway2
LargeAirway2
Trachea
Macrophage2

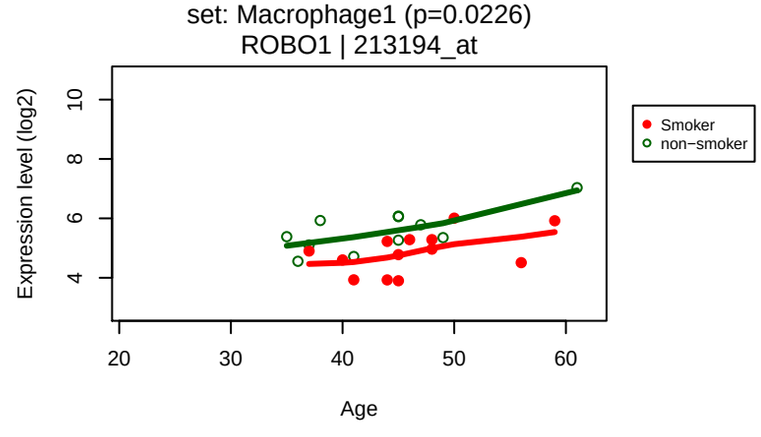

set: SmallAirway1 (p=0.1227)
ROBO1 | 213194_at
Expression level (log2)
Age
Smoker
non−smoker

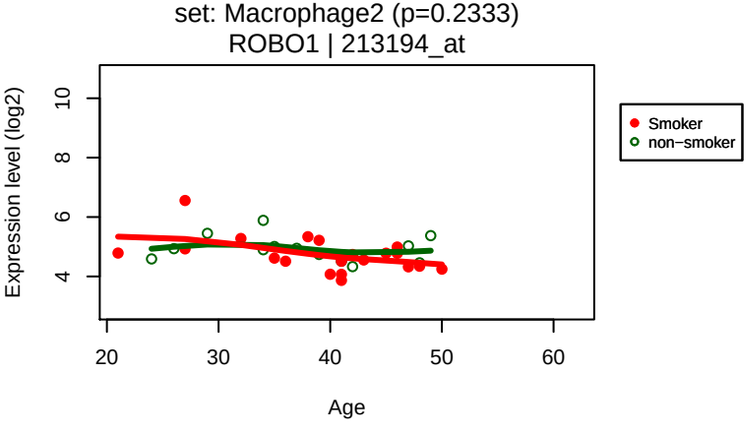

set: SmallAirway2 (p=0.0777)
ROBO1 | 213194_at
Expression level (log2)
Age
Smoker
non−smoker

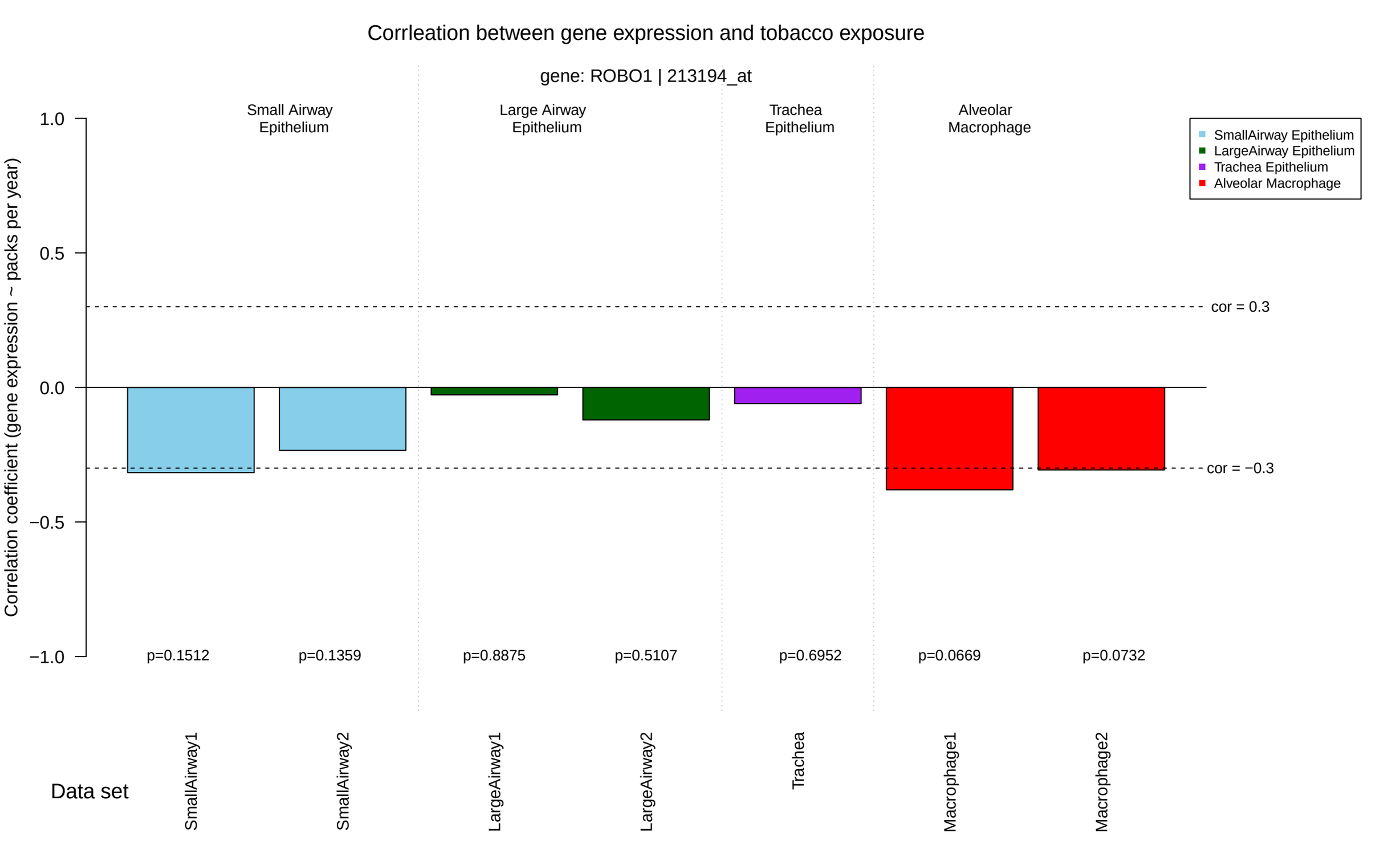

set: LargeAirway1 (p=0.0054)
ROBO1 | 213194_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0533)
ROBO1 | 213194_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7877)
ROBO1 | 213194_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0226)
ROBO1 | 213194_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2333)
ROBO1 | 213194_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ROBO1 | 213194_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1512
p=0.1359
p=0.8875
p=0.5107
p=0.6952
p=0.0669
p=0.0732
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2